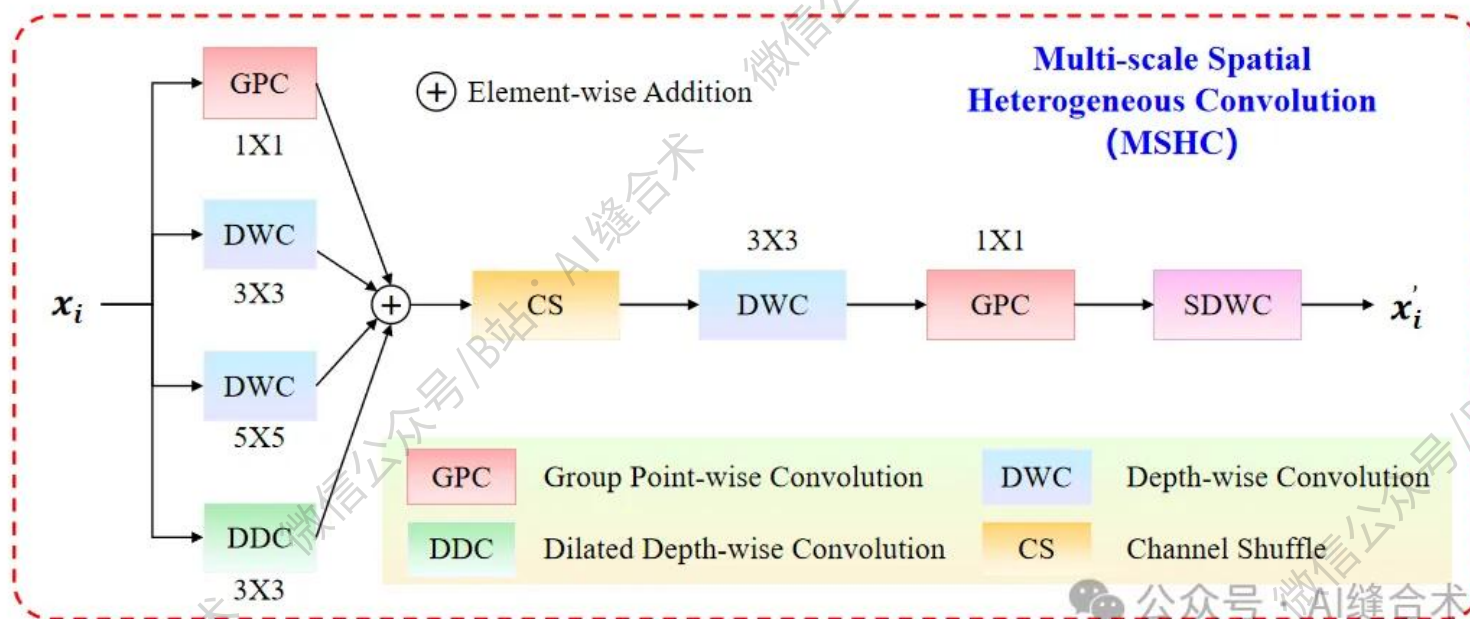


核心速览



多尺度空间异构卷积 (MSHC) :

通过异构卷积核 + 多尺度并行分支 + 通道混洗的组合，在大量使用轻量卷积（深度卷积、分组卷积）控制参数量和计算量的同时，大幅提升了多尺度空间特征的表达能力，尤其适合轻量级语义分割、目标检测等视觉任务。

<https://mp.weixin.qq.com/s/tyukr2lee1Ql-sxdn1a> PA

```

MSHC(
  (branch_gpc): Sequential(
    (0): Conv2d(32, 32, kernel_size=(1, 1), stride=(1, 1), groups=4, bias=False)
    (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU(inplace=True)
  )
  (branch_dwc3): Sequential(
    (0): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=32, bias=False)
    (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU(inplace=True)
  )
  (branch_dwc5): Sequential(
    (0): Conv2d(32, 32, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=32, bias=False)
    (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU(inplace=True)
  )
  (branch_ddc): Sequential(
    (0): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(2, 2), dilation=(2, 2), groups=32, bias=False)
    (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU(inplace=True)
  )
  (channel_shuffle): ChannelShuffle()
  (post_process): Sequential(
    (0): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=32, bias=False)
    (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU(inplace=True)
    (3): Conv2d(32, 32, kernel_size=(1, 1), stride=(1, 1), groups=4, bias=False)
    (4): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (5): ReLU(inplace=True)
    (6): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=32, bias=False)
    (7): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  )
)
input_tensor_shape : torch.Size([2, 32, 256, 256])
output_tensor_shape : torch.Size([2, 32, 256, 256])

```

哔哩哔哩/微信公众号：AI缝合术，独家整理！